

Lab 1: Array Reversal
Advanced Computerarchitecture

(B-KUL-JLI1QS)

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This study serves as an initiation into the realm of CUDA and GPU programming, exploring the efficiency of array reversal operations on both an NVIDIA MX250 graphics processing unit (GPU) and an Intel(R) Core(TM) i7-1065G7 CPU @ 1.30GHz. The Intel CPU boasts a clock speed of 1498 Mhz, housing 4 physical cores and 8 logical processors.

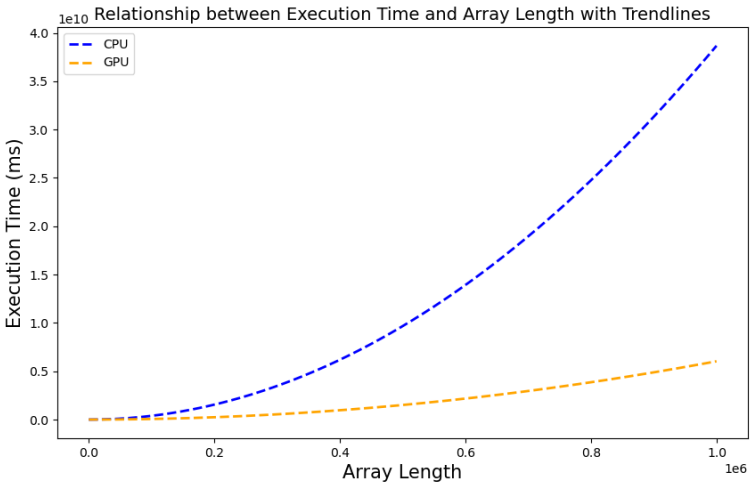


Figure 1: Execution Times in function of Array Length

For small array sizes, up to the order of 10^3, the CPU demonstrates lower execution times compared to the GPU. This difference is attributed to the substantial time overhead associated with thread spawning in relation to the reversal operation for smaller arrays. However, as illustrated in Figure 1, a notable trend emerges for larger array sizes, wherein the GPU consistently outperforms the CPU. Importantly, this favorable performance trajectory remains consistent across various configurations, demonstrating a consistent superiority of the GPU over the CPU as array sizes increase.

It is crucial to acknowledge the architectural limitations. A constraint exists on the number of threads per block, as all threads of a block are expected to reside on the same streaming multiprocessor core and must share the limited memory resources of that core. On current GPUs, a thread block may contain up to 1024 threads. However, a kernel can be executed by multiple equally-shaped thread blocks, resulting in the total number of threads being equal to the number of threads per block multiplied by the number of blocks.

Moreover, the investigation conducted thorough experimentation with a spectrum of thread configurations, encompassing diverse counts of threads per block and blocks per grid. This exhaustive exploration yielded valuable insights into the complexities of CUDA and GPU programming, delivering a nuanced comprehension of how different configurations influence the performance of programs.

The GitHub repository containing the source code can be found at *Github Repository*